

Inequalities

- **Exercise 1**

Let $f(x) = \frac{2x-1}{x+3}$. Find the preimage $f^{-1}((1, 3))$ of $(1, 3)$.

- **Solution 1.1**

We have

$$\begin{aligned} f^{-1}((1, 3)) &= \{x \in \mathbb{R} \setminus \{-3\} : f(x) \in (1, 3)\} \\ &= \left\{x \in \mathbb{R} \setminus \{-3\} : 1 < \frac{2x-1}{x+3} < 3\right\} \\ &= \left\{x \in \mathbb{R} \setminus \{-3\} : 1 < \frac{2x-1}{x+3}\right\} \cap \left\{x \in \mathbb{R} \setminus \{-3\} : \frac{2x-1}{x+3} < 3\right\} \\ &= \left\{x \in \mathbb{R} \setminus \{-3\} : 0 < \frac{2x-1}{x+3} - 1\right\} \cap \left\{x \in \mathbb{R} \setminus \{-3\} : \frac{2x-1}{x+3} - 3 < 0\right\} \\ &= \left\{x \in \mathbb{R} \setminus \{-3\} : 0 < \frac{x-4}{x+3}\right\} \cap \left\{x \in \mathbb{R} \setminus \{-3\} : \frac{-x-10}{x+3} < 0\right\} \\ &= ((-\infty, -3) \cup (4, \infty)) \cap ((-\infty, -10) \cup (-3, \infty)) \\ &= (-\infty, -10) \cup (4, \infty). \end{aligned}$$

- **Exercise 2**

Suppose $x, y \in \mathbb{R}$ satisfy

$$0 \leq |x - y| < \varepsilon$$

for every $\varepsilon > 0$. Prove that $x = y$.

- **Proof 2.1**

Suppose $x, y \in \mathbb{R}$. Consider $a = |x - y|$. Then $\forall \varepsilon > 0, a \geq 0, a < \varepsilon$. By Theorem 1.5, we have that $a = 0 = |x - y|$, that is, $x = y$.

- **Exercise 3**

Suppose $a \in \mathbb{R}$ satisfies

$$0 \leq a < \frac{1}{n}$$

for every $n \in \mathbb{N}$. Prove that $a = 0$.

- **Proof 3.1**

Suppose $a \in \mathbb{R}$. Suppose $\forall n \in \mathbb{N}, 0 \leq a < \frac{1}{n}$. Suppose $\varepsilon > 0$, we need to prove that $a < \varepsilon$. By the Archimedean property, $\exists N_\varepsilon \in \mathbb{N}$ such that $\frac{1}{N_\varepsilon} < \varepsilon$. Thus $0 \leq a < \frac{1}{N_\varepsilon} < \varepsilon$. Therefore $0 \leq a < \varepsilon$. By Theorem 1.5, we have that $a = 0$.

- **Exercise 4**

Suppose $a \in \mathbb{R}$, suppose that for all $\varepsilon > 0, a < \varepsilon$. Is it true that $a = 0$? Prove it or give a counterexample.

- **Solution 4.1**

It's false. Consider $a = -1$, then $a < \varepsilon$ holds for all $\varepsilon > 0$, but $a \neq 0$.

- **Exercise 5**

Consider $f(x) = \frac{2x+1}{x-5}$. Find the elements in the domain whose image is at a distance less than 2 from 3.

- **Solution 5.1**

We have that

$$\begin{aligned}
\{x \in \mathbb{R} \setminus \{5\} : d(f(x), 3) < 2\} &= \left\{x \in \mathbb{R} \setminus \{5\} : \left| \frac{2x+1}{x-5} - 3 \right| < 2 \right\} \\
&= \left\{x \in \mathbb{R} \setminus \{5\} : \left| \frac{2x+1-3(x-5)}{x-5} \right| < 2 \right\} \\
&= \left\{x \in \mathbb{R} \setminus \{5\} : \left| \frac{-x+16}{x-5} \right| < 2 \right\}.
\end{aligned}$$

By Proposition 2.3(1), this is equivalent to:

$$-2 < \frac{-x+16}{x-5} < 2.$$

We solve this by splitting it into a system of two inequalities

1. For

$$\begin{aligned}
-2 < \frac{-x+16}{x-5} &\Leftrightarrow \frac{-x+16+2(x-5)}{x-5} > 0 \\
&\Leftrightarrow \frac{x+6}{x-5} > 0 \\
&\Rightarrow x \in (-\infty, -6) \cup (5, \infty).
\end{aligned}$$

2. For

$$\begin{aligned}
\frac{-x+16}{x-5} < 2 &\Leftrightarrow \frac{-x+16-2(x-5)}{x-5} < 0 \Leftrightarrow \frac{-3x+26}{x-5} < 0 \\
&\Leftrightarrow \frac{3x-26}{x-5} > 0 \\
&\Rightarrow x \in (-\infty, 5) \cup \left(\frac{26}{3}, \infty\right).
\end{aligned}$$

Taking the intersection of these two solution sets yields

$$((-\infty, -6) \cup (5, \infty)) \cap \left((-\infty, 5) \cup \left(\frac{26}{3}, \infty\right)\right) = (-\infty, -6) \cup \left(\frac{26}{3}, \infty\right).$$

Therefore, the required elements in the domain are $x \in (-\infty, -6) \cup \left(\frac{26}{3}, \infty\right)$.

• Exercise 6

Solve $|2x-5| - \left|\frac{x+1}{x-3}\right| \geq 1$.

o Solution 6.1

Consider a necessary condition $|2x-5| - 1 \geq 0$, which yields $2x-5 \leq -1 \vee 2x-5 \geq 1 \Leftrightarrow x \leq 2 \vee x \geq 3$. Notice that the expression is undefined at $x=3$, so the domain is $\mathbb{R} \setminus \{3\}$. The zeroes of the absolute value expressions are $x = \frac{5}{2}$, $x = -1$, and $x = 3$. Combined with the necessary condition, we consider the following valid intervals for x :

1. **Case 1:** $x \leq -1$.

In this interval, $2x-5 < 0 \Rightarrow |2x-5| = -2x+5$. For the fraction, $x+1 \leq 0$ and $x-3 < 0$, so $\frac{x+1}{x-3} \geq 0 \Rightarrow \left|\frac{x+1}{x-3}\right| = \frac{x+1}{x-3}$. The inequality becomes

$$-2x+5 - \frac{x+1}{x-3} \geq 1 \Leftrightarrow -2x+4 - \frac{x+1}{x-3} \geq 0 \Leftrightarrow \frac{(-2x+4)(x-3) - (x+1)}{x-3} \geq 0.$$

Expanding the numerator: $-2x^2 + 10x - 12 - x - 1 = -2x^2 + 9x - 13$. The inequality is

$$\frac{-2x^2 + 9x - 13}{x-3} \geq 0 \Leftrightarrow \frac{2x^2 - 9x + 13}{x-3} \leq 0.$$

The discriminant of the quadratic equation $2x^2 - 9x + 13$ is $\Delta = (-9)^2 - 4(2)(13) = 81 - 104 = -23 < 0$, which means $2x^2 - 9x + 13 > 0$ for all $x \in \mathbb{R}$. Thus, the inequality depends only on $x-3 < 0 \Rightarrow x < 3$. Intersecting with $x \leq -1$ yields $x \in (-\infty, -1]$.

2. **Case 2:** $-1 < x \leq 2$.

In this interval, $|2x - 5| = -2x + 5$. For the fraction, $x + 1 > 0$ and $x - 3 < 0$, so $\frac{x+1}{x-3} < 0$
 $\Rightarrow \left| \frac{x+1}{x-3} \right| = -\frac{x+1}{x-3}$. The inequality becomes

$$-2x + 5 + \frac{x+1}{x-3} \geq 1 \iff -2x + 4 + \frac{x+1}{x-3} \geq 0 \iff \frac{(-2x+4)(x-3) + (x+1)}{x-3} \geq 0.$$

Expanding the numerator $-2x^2 + 10x - 12 + x + 1 = -2x^2 + 11x - 11$. The inequality is:

$$\frac{-2x^2 + 11x - 11}{x-3} \geq 0 \iff \frac{2x^2 - 11x + 11}{x-3} \leq 0.$$

Since $x \leq 2 < 3$, the denominator $x - 3$ is negative, so the numerator must be positive, that is, $2x^2 - 11x + 11 \geq 0$.

The roots of $2x^2 - 11x + 11 = 0$ are $x = \frac{11 \pm \sqrt{121 - 88}}{4} = \frac{11 \pm \sqrt{33}}{4}$. So The solution for the numerator is $x \leq \frac{11 - \sqrt{33}}{4} \vee x \geq \frac{11 + \sqrt{33}}{4}$.

Intersecting this with the case interval $-1 < x \leq 2$ gives $x \in \left(-1, \frac{11 - \sqrt{33}}{4} \right]$.

3. **Case 3:** $x > 3$.

In this interval, $2x - 5 > 0 \Rightarrow |2x - 5| = 2x - 5$. For the fraction, $x + 1 > 0$ and $x - 3 > 0$, so $\left| \frac{x+1}{x-3} \right| = \frac{x+1}{x-3}$. The inequality becomes

$$2x - 5 - \frac{x+1}{x-3} \geq 1 \iff 2x - 6 - \frac{x+1}{x-3} \geq 0 \iff \frac{2(x-3)^2 - (x+1)}{x-3} \geq 0.$$

Since $x > 3$, the denominator is positive, so the numerator must be positive, that is, $2x^2 - 12x + 18 - x - 1 = 2x^2 - 13x + 17 \geq 0$.

The roots are $x = \frac{13 \pm \sqrt{169 - 136}}{4} = \frac{13 \pm \sqrt{33}}{4}$. So the solution is $x \leq \frac{13 - \sqrt{33}}{4} \vee x \geq \frac{13 + \sqrt{33}}{4}$.

Intersecting this with the case interval $x > 3$ gives $x \in \left[\frac{13 + \sqrt{33}}{4}, \infty \right)$.

Combining all three cases, the complete solution set is:

$$(-\infty, -1] \cup \left(-1, \frac{11 - \sqrt{33}}{4} \right] \cup \left[\frac{13 + \sqrt{33}}{4}, \infty \right) = \left(-\infty, \frac{11 - \sqrt{33}}{4} \right] \cup \left[\frac{13 + \sqrt{33}}{4}, \infty \right).$$

• **Exercise 7**

Prove that $\forall \varepsilon > 0, \exists \delta > 0$, such that $\forall x, 0 < |x + 3| < \delta \Rightarrow |2x + 6| < \varepsilon$.

◦ **Proof 7.1**

Suppose $\varepsilon > 0$ and let $\delta := \frac{\varepsilon}{2}$. Suppose $0 < |x + 3| < \delta$. Then we have:

$$|2x + 6| = 2|x + 3| < 2\delta = 2\left(\frac{\varepsilon}{2}\right) = \varepsilon.$$

Therefore, the implication holds.

• **Exercise 8**

Prove that $\forall \varepsilon > 0, \exists \delta > 0$, such that $\forall x, 0 < |x - 2| < \delta \Rightarrow |x^3 - 8| < \varepsilon$.

◦ **Proof 8.1**

Suppose $\varepsilon > 0$ and let $\delta := \min \left\{ 1, \frac{\varepsilon}{19} \right\}$. Suppose $0 < |x - 2| < \delta$.

Since $\delta \leq 1$, we have $|x - 2| < 1$, which implies:

$$-1 < x - 2 < 1 \implies 1 < x < 3.$$

We factor the expression $|x^3 - 8|$ as $|x - 2| \cdot |x^2 + 2x + 4|$. To bound the term $|x^2 + 2x + 4|$, we apply the triangle inequality given $1 < x < 3$:

$$|x^2 + 2x + 4| \leq |x|^2 + 2|x| + 4 < 3^2 + 2(3) + 4 = 19.$$

Since we also have $|x - 2| < \delta \leq \frac{\varepsilon}{19}$, it follows that:

$$|x^3 - 8| = |x - 2| \cdot |x^2 + 2x + 4| < \left(\frac{\varepsilon}{19}\right) \cdot 19 = \varepsilon.$$

Therefore, the statement is proved.

• **Exercise 9**

Prove that $\forall \varepsilon > 0, \exists \delta > 0$, such that $\forall x, 0 < |x - 1| < \delta \implies \left| \frac{1}{x^2} - 1 \right| < \varepsilon$.

◦ **Proof 9.1**

Suppose $\varepsilon > 0$ and let $\delta := \min \left\{ \frac{1}{2}, \frac{\varepsilon}{10} \right\}$. Suppose $0 < |x - 1| < \delta$.

Since $\delta \leq \frac{1}{2}$, we have $|x - 1| < \frac{1}{2}$, which implies:

$$-\frac{1}{2} < x - 1 < \frac{1}{2} \implies \frac{1}{2} < x < \frac{3}{2}.$$

From $\frac{1}{2} < x$, we have $x^2 > \frac{1}{4} \implies \frac{1}{x^2} < 4$. Also, from $x < \frac{3}{2}$, we have $|x + 1| \leq |x| + 1 < \frac{3}{2} + 1 = \frac{5}{2}$.

Now, we algebraically simplify and factor the target expression:

$$\left| \frac{1}{x^2} - 1 \right| = \left| \frac{1 - x^2}{x^2} \right| = \frac{|1 - x| \cdot |1 + x|}{x^2} = |x - 1| \cdot \frac{|x + 1|}{x^2}.$$

Using our bounds $\frac{1}{x^2} < 4$ and $|x + 1| < \frac{5}{2}$, we obtain:

$$\frac{|x + 1|}{x^2} < 4 \cdot \frac{5}{2} = 10.$$

Since we also have $|x - 1| < \delta \leq \frac{\varepsilon}{10}$, it follows that:

$$\left| \frac{1}{x^2} - 1 \right| = |x - 1| \cdot \frac{|x + 1|}{x^2} < \left(\frac{\varepsilon}{10}\right) \cdot 10 = \varepsilon.$$

Therefore, the statement is proved.